Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

Roll Number: 2520030105
Name: KARKALA SHIVA REDDY
Section -3

Cloud Infrastructure and Services

25CS2206E

PRACTICAL RECORD

2026

Experiment -2

AIM: To enable Multi-Factor Authentication (MFA) for secure access to the AWS account, create a first Linux instance in Amazon EC2 and connect to it, and create an S3 bucket and upload five objects with different file extensions.

Description:

Multi-Factor Authentication (MFA) adds a second verification step to the normal user name and password. After MFA is enabled, the account can be accessed only when the password and the second factor are both provided. AWS supports passkeys and security keys as MFA devices, which use the fingerprint, face unlock or screen lock of the device and give strong protection against phishing and password theft. Since the root user has full access to the account, enabling MFA for the root user is the first security step.

Amazon Elastic Compute Cloud (EC2) provides resizable computing capacity in the cloud. A Linux instance is created from an Amazon Machine Image (AMI), which contains the operating system and the software required to start the instance. The Amazon Linux 2023 AMI is free tier eligible and the t3.micro instance type can be used without charges within the free tier limits. A key pair and a security group are selected while launching, and the instance can be connected from the browser using EC2 Instance Connect. The instance is terminated after the work is completed so that no charges are incurred.

Amazon Simple Storage Service (S3) is an object storage service in which data is stored as objects inside buckets. Every bucket name must be unique and is created in a selected AWS Region. Block Public Access keeps the bucket private, and objects are added to the bucket using the Upload option. In this practical, one bucket was created and five objects with different file extensions were uploaded to it, and a second bucket was created to understand unique bucket naming in the global namespace.

Steps:

a) Enabling Multi-Factor Authentication

1. Sign in to the AWS Management Console with the root account and open the IAM service.

2. In My security credentials, scroll to the Multi-factor authentication (MFA) section and click Assign MFA device.

3. Enter a device name, select the Passkey or security key option and continue.

4. Complete the browser prompt and save the passkey in Google Password Manager.

5. Confirm that the MFA table lists the new passkey device with the current date.

b) Creating First Linux Instance

6. Type EC2 in the console search bar and open the EC2 service.

7. On the EC2 Dashboard, click Launch instance.

8. Enter the instance name as MYCOMPUTER.

9. In Application and OS Images, select the Amazon Linux 2023 AMI (64-bit x86, free tier eligible).

10. Keep the instance type as t3.micro.

11. Select or create a key pair for logging in to the instance.

12. Keep the default VPC and create a new security group that allows SSH access.

13. Leave the storage at 8 GiB gp3 and the advanced details at their default values.

14. Review the summary and click Launch instance.

15. Note the success message and the instance ID i-02a5f5fe0ebec9742.

16. Wait until the instance state is Running and the status check shows 3/3 checks passed.

17. Select the instance, click Connect and choose EC2 Instance Connect with the public IPv4 address.

18. In the browser terminal, run ls, mkdir test, ls, vi file.txt, cat file.txt, history and grep to practise basic Linux commands.

19. Terminate the instance after the commands are verified, so that no charges are incurred.

c) S3 Bucket Creation and Upload Objects

20. Type S3 in the console search bar and open Amazon S3.

21. Click Create bucket and select the Account Regional namespace.

22. Enter the bucket name shiva-s3-bucket-741973715162-eu-north-1-an and select the Europe (Stockholm) region.

23. Keep ACLs disabled and keep Block all public access enabled for the first bucket.

24. Leave versioning disabled and add no tags, then click Create bucket.

25. Open the bucket, click Upload and add five files with different extensions

(.pdf, .txt, .jpg, .csv and .docx).

26. Keep the default permissions and Standard storage class, then click Upload.

27. Verify the uploaded objects in the Objects tab of the bucket.

28. Create a second bucket shiva-s3-bucket1-741973715162-eu-north-1-an using the Global namespace and review its public access settings.

29. Open the Buckets page and confirm that both buckets are listed.

Result: Thus, Multi-Factor Authentication was enabled for the AWS account using a passkey, a first Linux instance was launched, connected and terminated successfully, and an S3 bucket was created with five objects of different file extensions. A second bucket was also created to verify unique bucket naming.

Output:

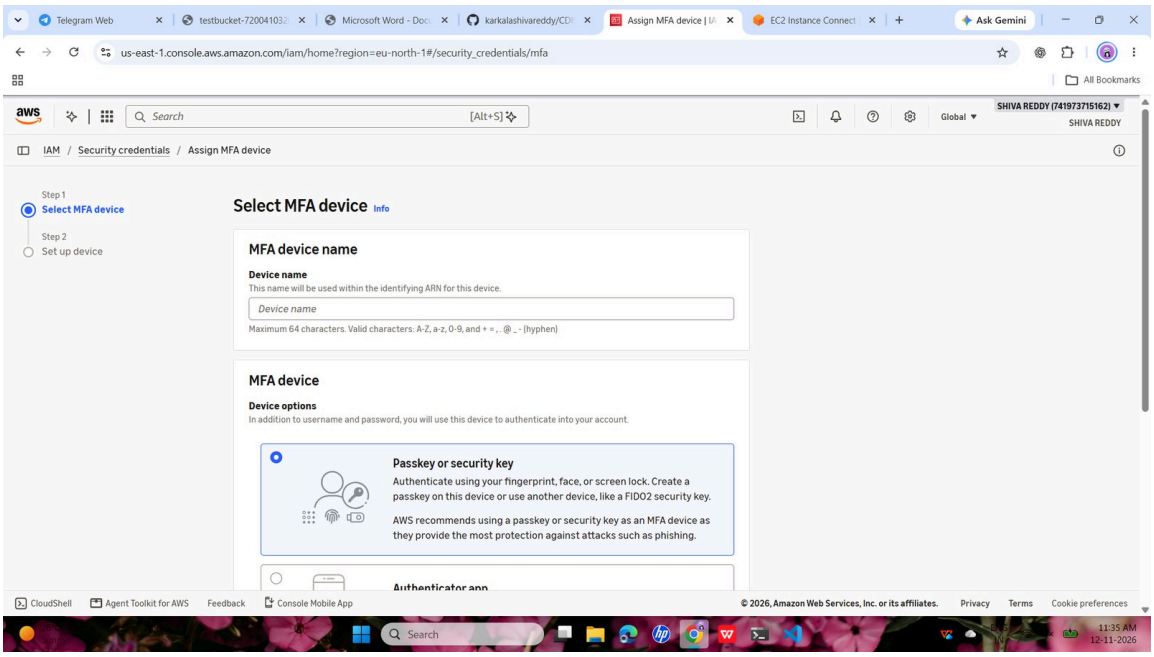


Fig 1: Assign MFA device - selecting the device type

The Assign MFA device page was opened from the security credentials. The Passkey or security key option was selected, which lets the browser save a passkey for the account.

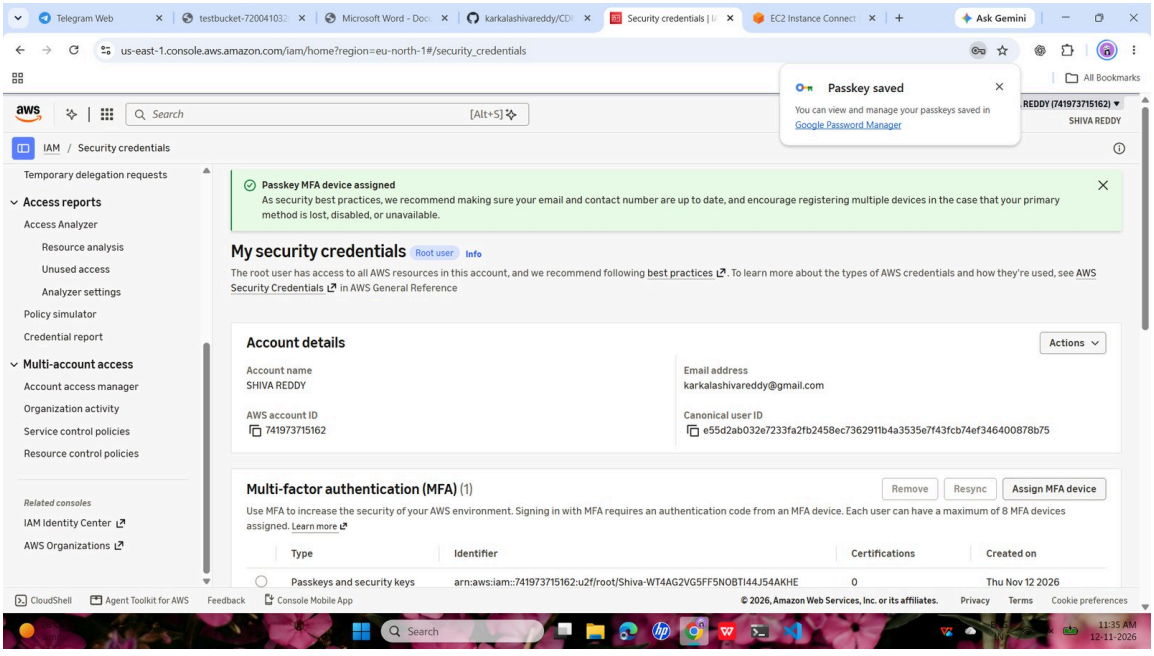


Fig 2: Passkey MFA device assigned successfully

The passkey was saved through Google Password Manager and the MFA device was assigned. The MFA table lists one passkey device created on 12 November 2026 for the root user, so a second factor is now required at sign-in.

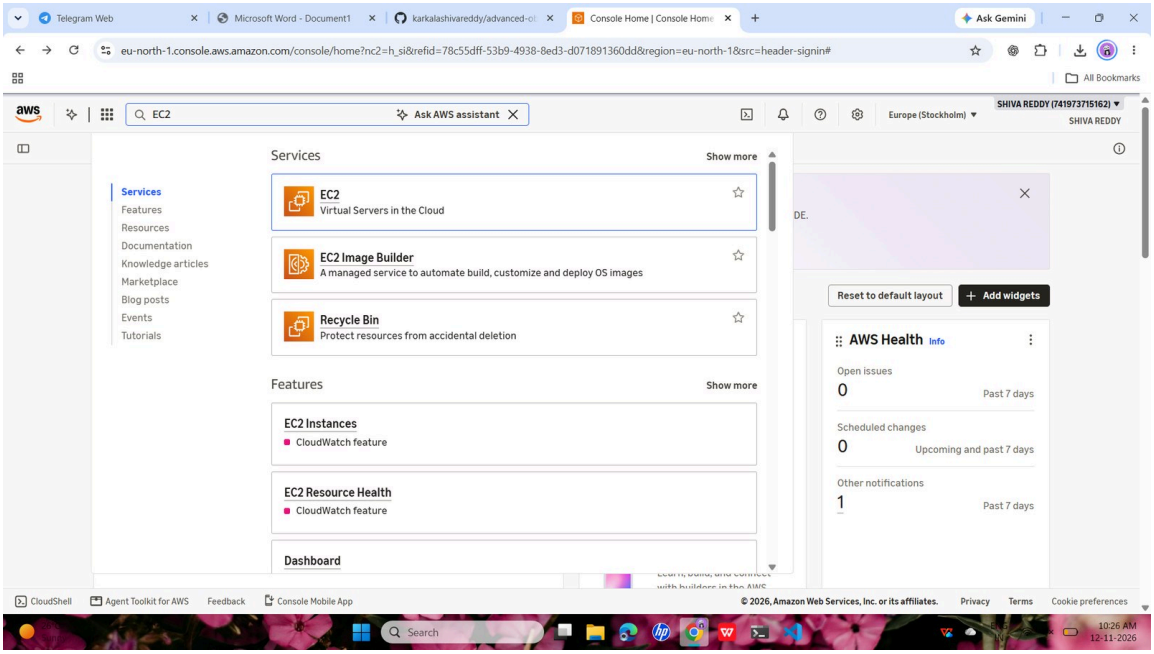


Fig 3: Searching for EC2 in the AWS console

The console home page was opened with the region set to Europe (Stockholm). EC2 was typed in the search bar and the EC2 service appeared at the top of the results, which was opened to start the instance.

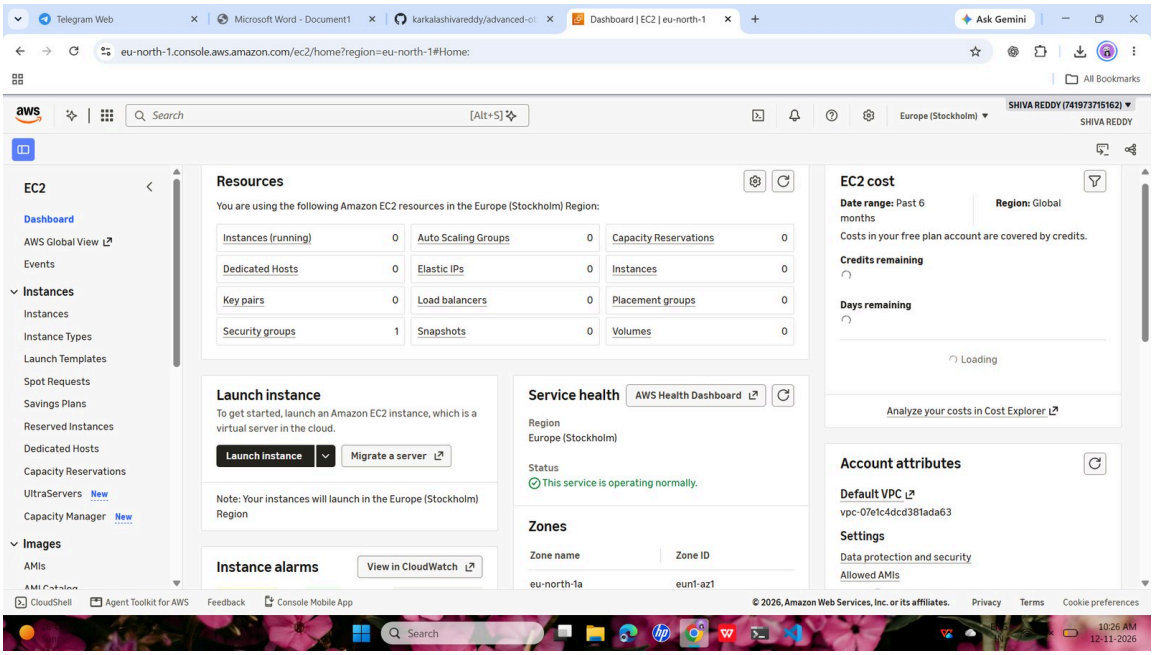


Fig 4: EC2 Dashboard before launching the instance

The EC2 Dashboard showed that no instances were running in the Stockholm region. The default VPC and one existing security group were present, and the Launch instance button was used to begin.

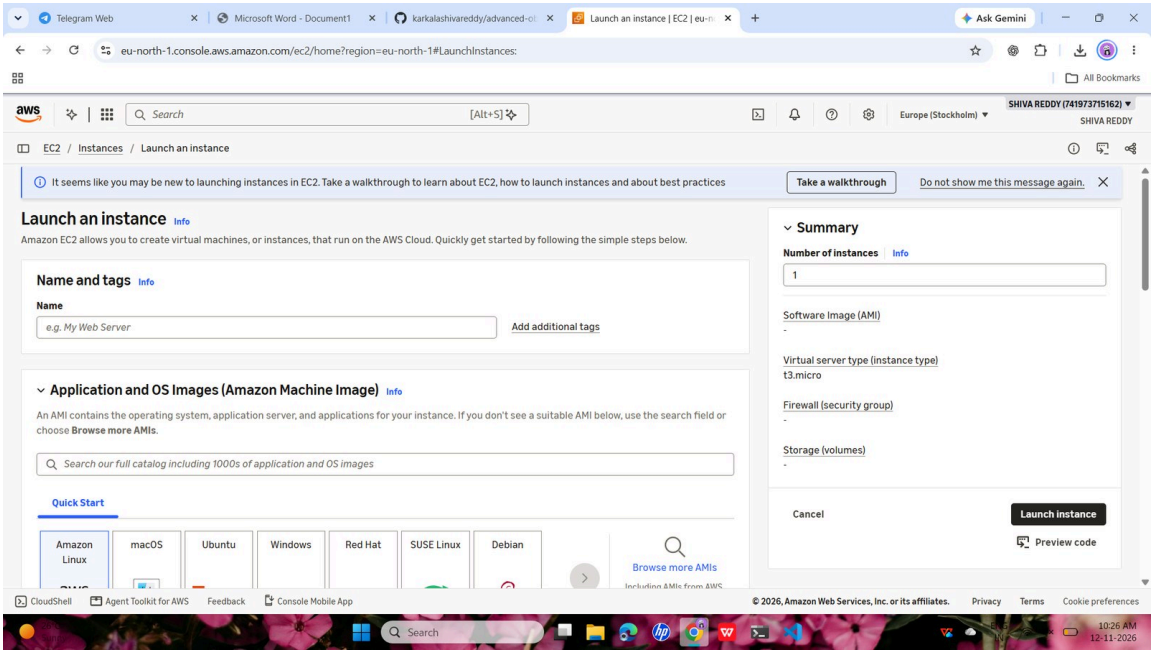


Fig 5: Launch instance - name and AMI selection

The launch wizard was opened with one instance and the t3.micro type selected by default. The instance name was entered and the Amazon Linux quick start tab was used to choose the operating system.

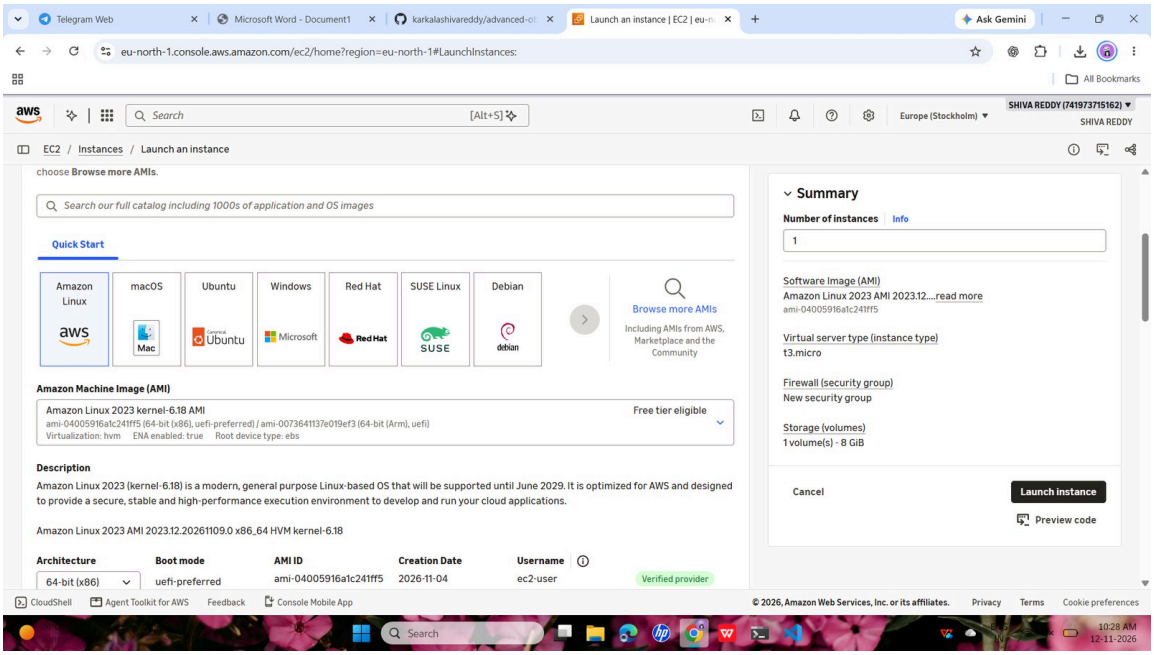


Fig 6: Amazon Linux 2023 AMI selected

The Amazon Linux 2023 AMI (64-bit x86, AMI ID ami-04005916a1c241ff5) was selected. It is free tier eligible, uses an EBS root volume and the default login user is ec2-user.

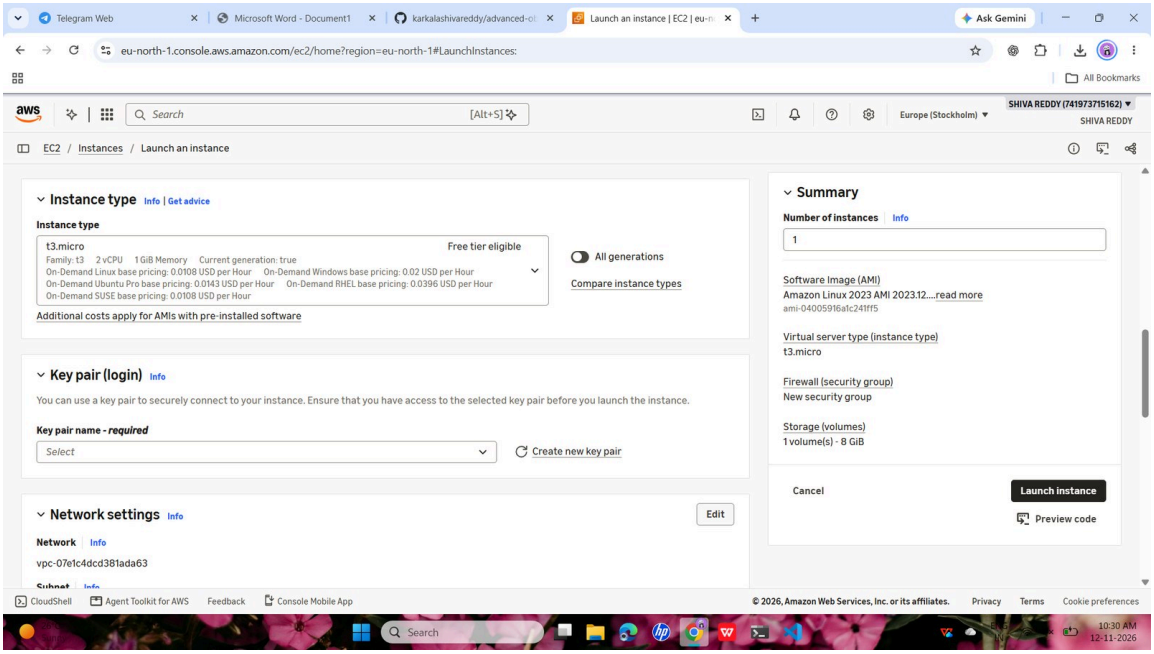


Fig 7: Instance type, key pair and network settings

The instance type was confirmed as t3.micro with 2 vCPU and 1 GiB memory. The key pair and network settings were configured in this section, using the default VPC and a new security group.

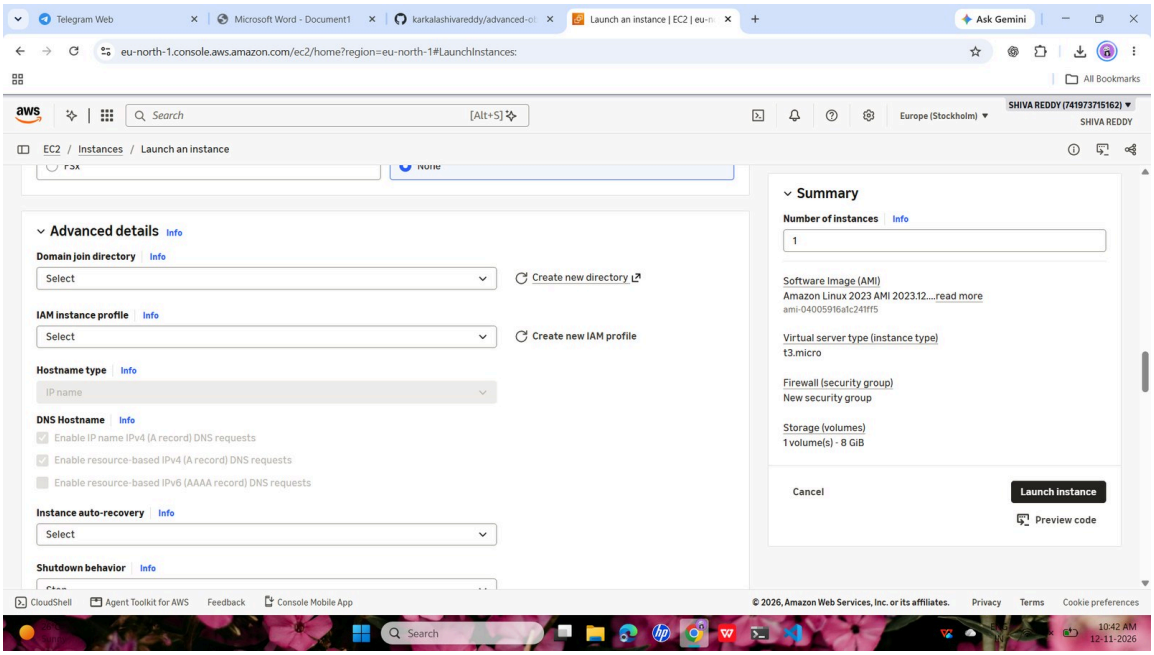


Fig 8: Advanced details of the launch

The advanced details were left at their default values. The hostname type was set to IP name with the IPv4 DNS options enabled, and no IAM instance profile was attached.

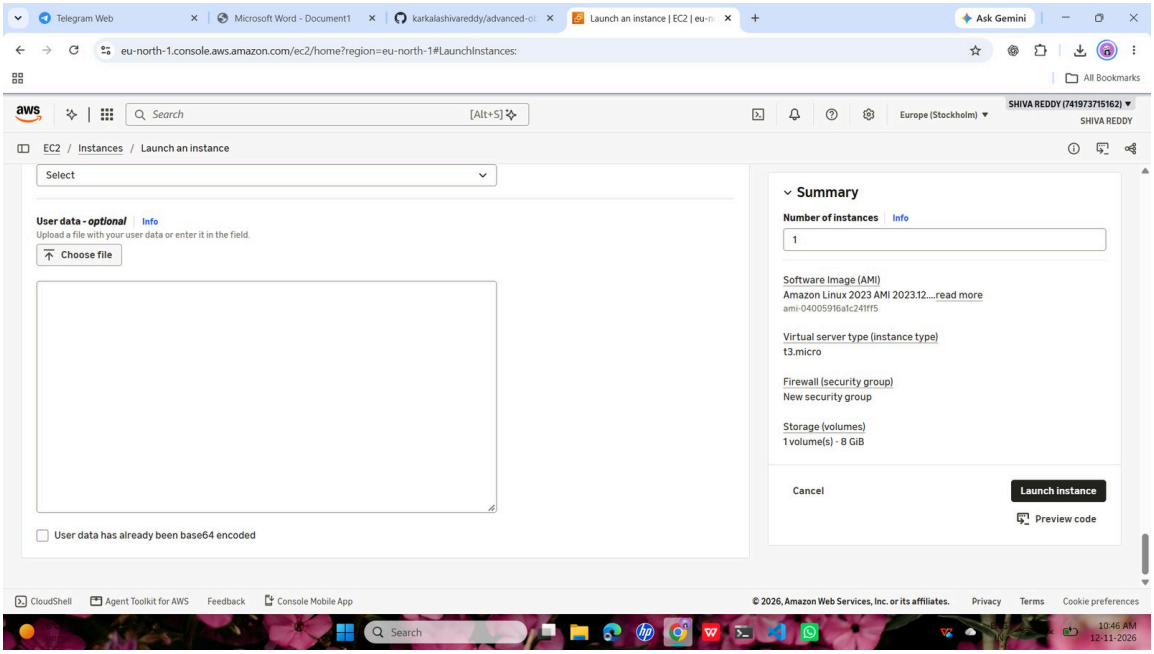


Fig 9: Reviewing the instance configuration

Before launching, the summary was checked. It showed one instance, the Amazon Linux 2023 AMI, the t3.micro type, a new security group and an 8 GiB root volume. The user data field was left empty.

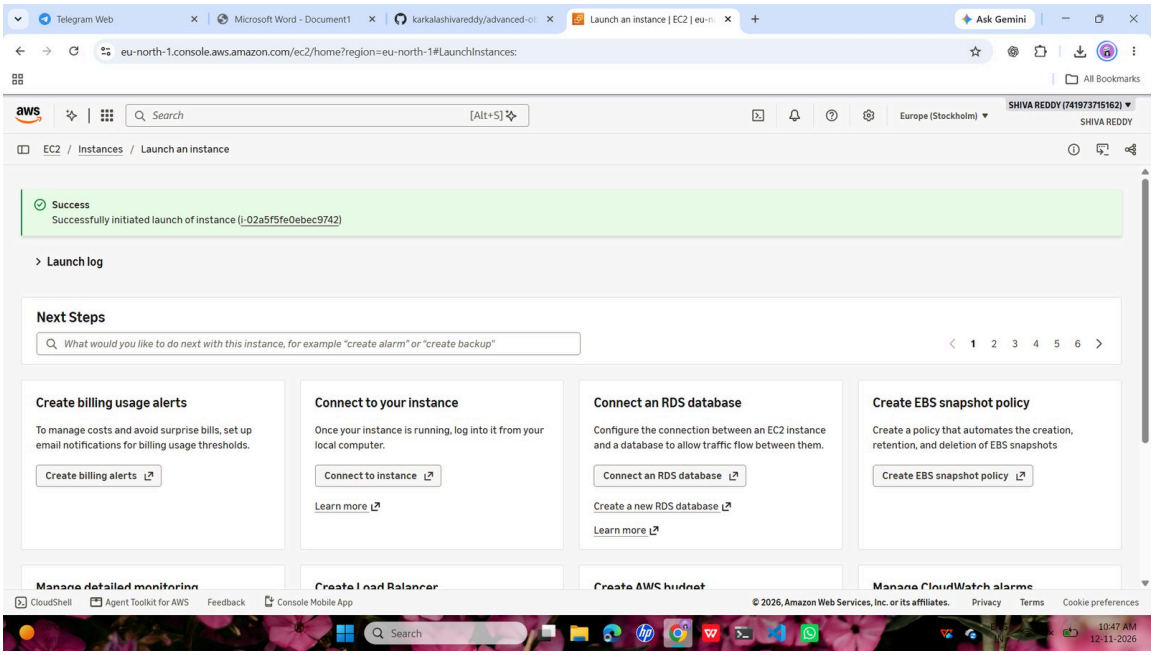


Fig 10: Launch success message

After clicking Launch instance, a green success message showed that the instance i-02a5f5fe0ebec9742 was launched. The Next Steps panel offered the option to connect to the instance.

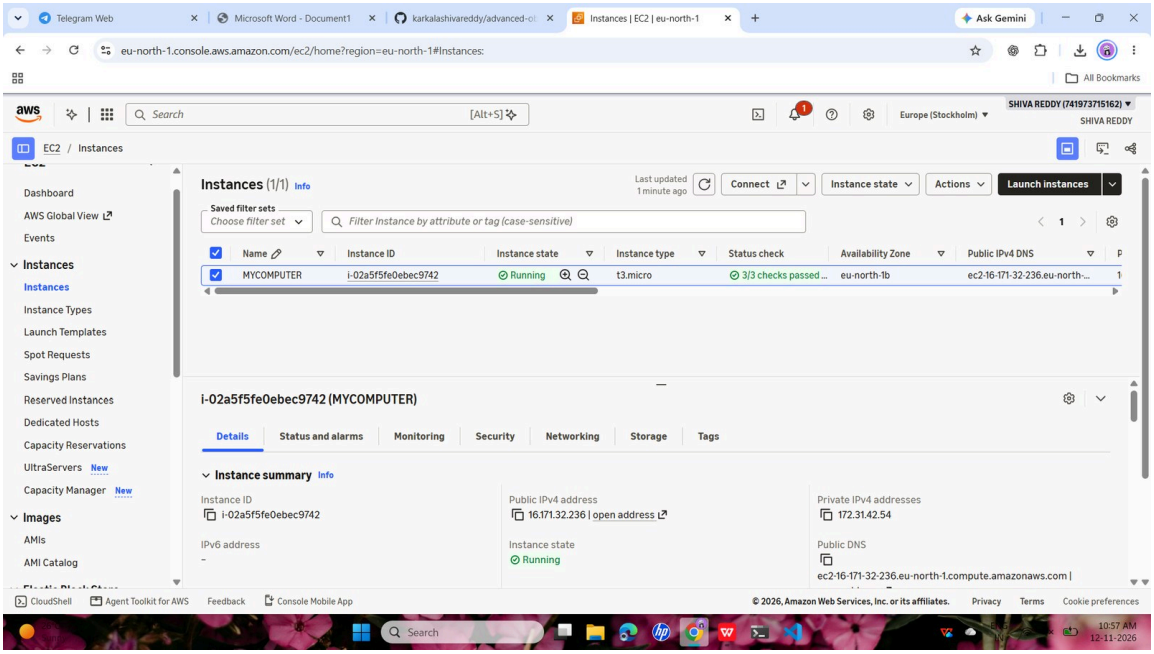


Fig 11: MYCOMPUTER instance running

The instance named MYCOMPUTER appeared in the Instances list with the state Running. It was running in eu-north-1b with public IP 16.171.32.236 and all 3/3 status checks passed.

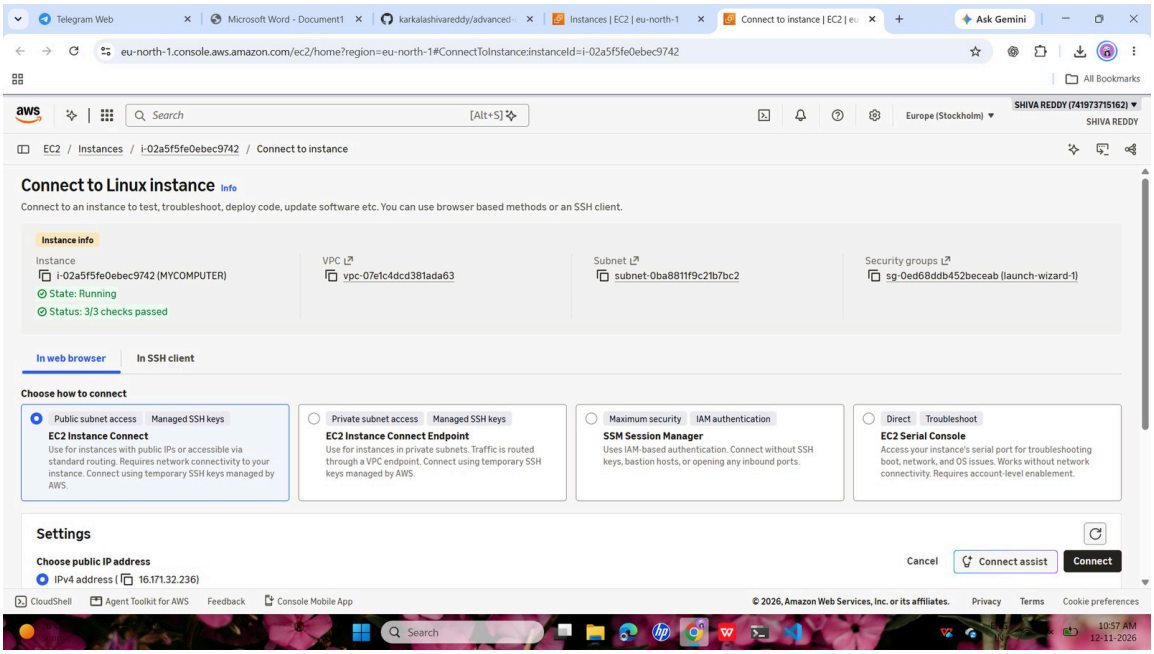


Fig 12: Connecting to the Linux instance

The Connect page was opened for the instance. The EC2 Instance Connect option was selected with the public IPv4 address, which opens a terminal in the browser without a separate SSH client.

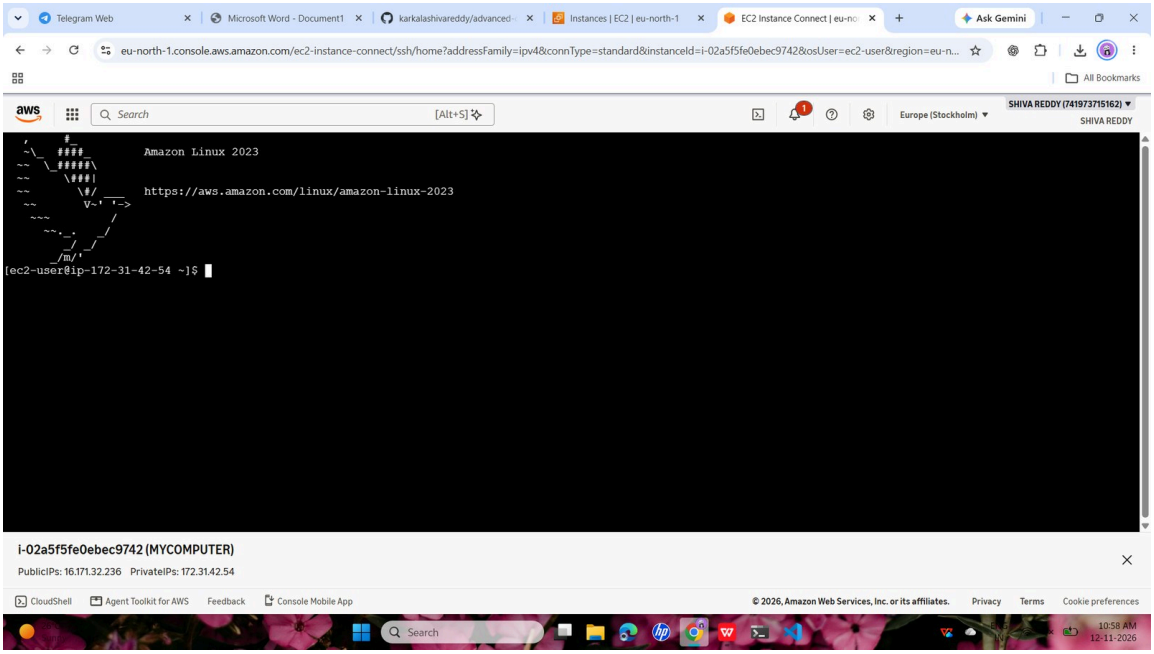


Fig 13: Amazon Linux 2023 terminal session

The browser terminal connected to the instance and showed the Amazon Linux 2023 banner with the prompt ec2-user@ip-172-31-42-54. The private and public IP addresses are shown at the bottom of the terminal.

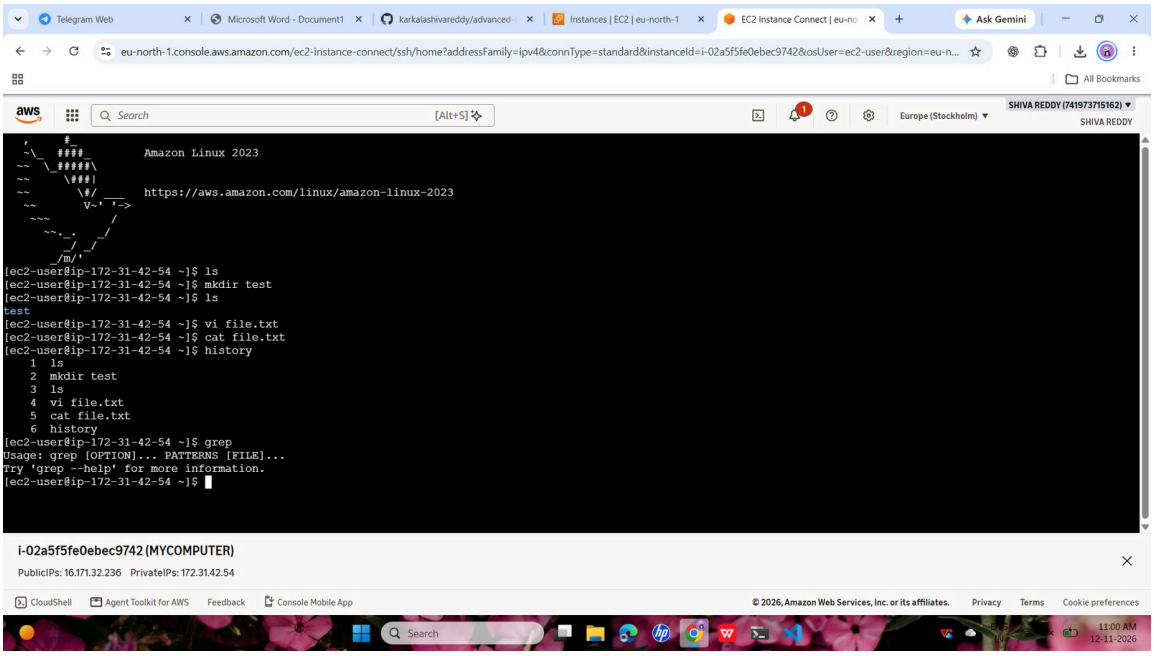


Fig 14: Linux commands executed on the instance

Some basic Linux commands were practised in the terminal. The ls, mkdir test, ls, vi file.txt, cat file.txt, history and grep commands were run to check the working of the Linux instance.

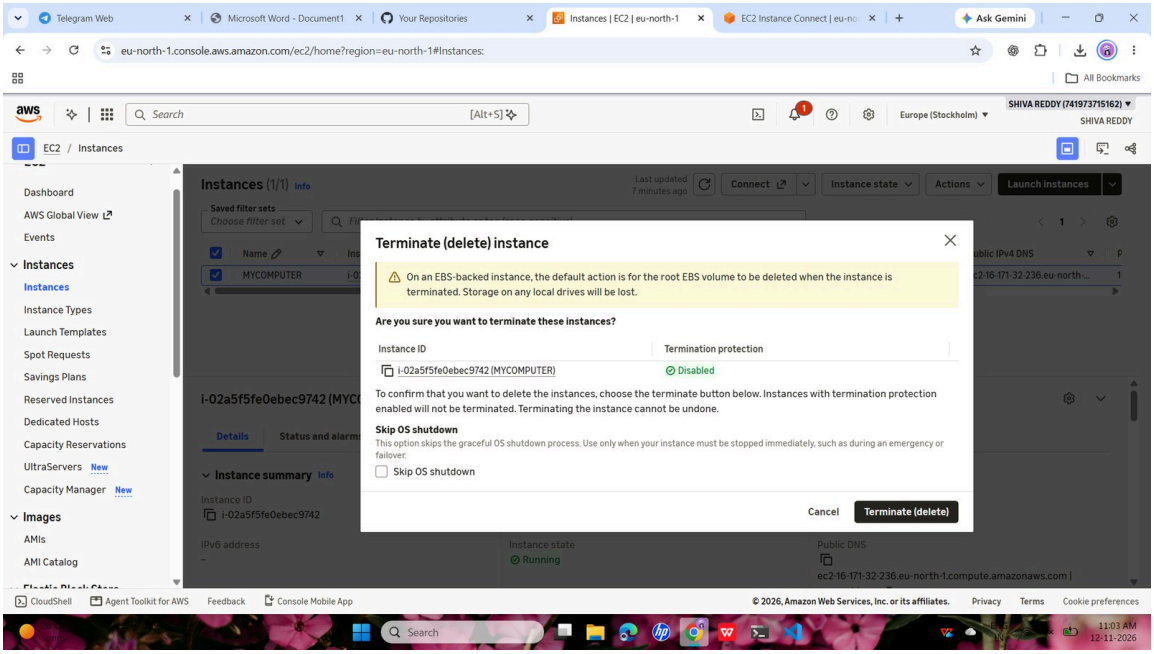


Fig 15: Terminate instance confirmation

After the commands were verified, the instance was terminated so that no charges are incurred. The confirmation dialog showed the instance ID and the warning that the root EBS volume is deleted on termination.

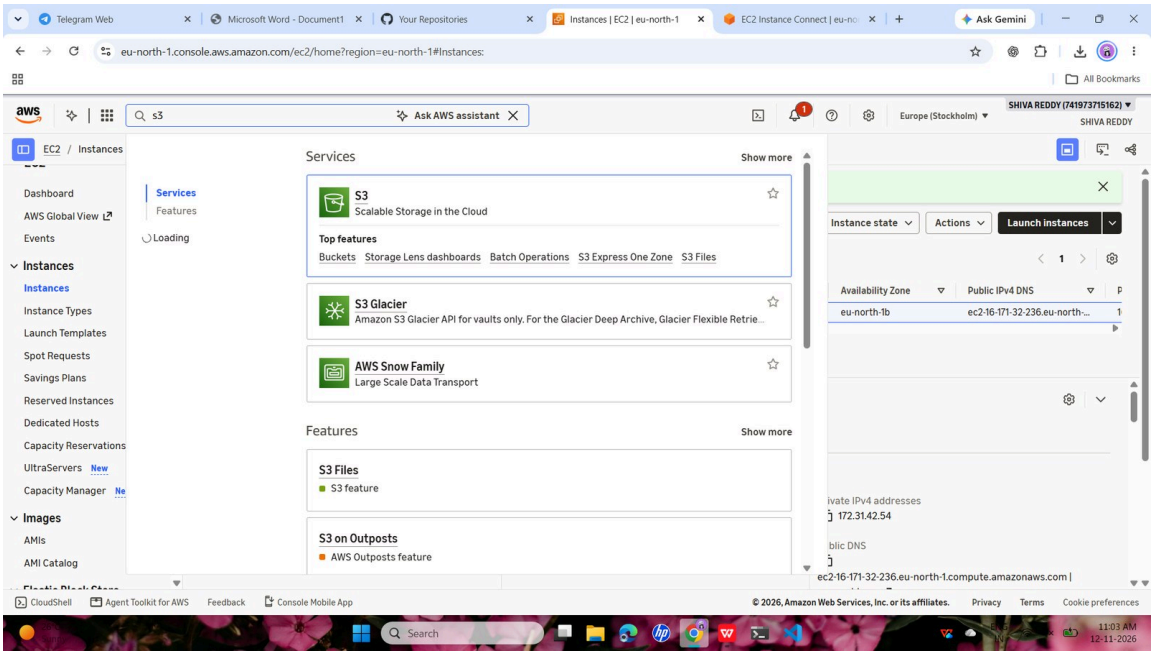


Fig 16: Searching for S3 in the console

Back in the console, S3 was typed in the search bar and the Amazon S3 service was opened from the results. The region remained Europe (Stockholm), so the buckets were created in eu-north-1.

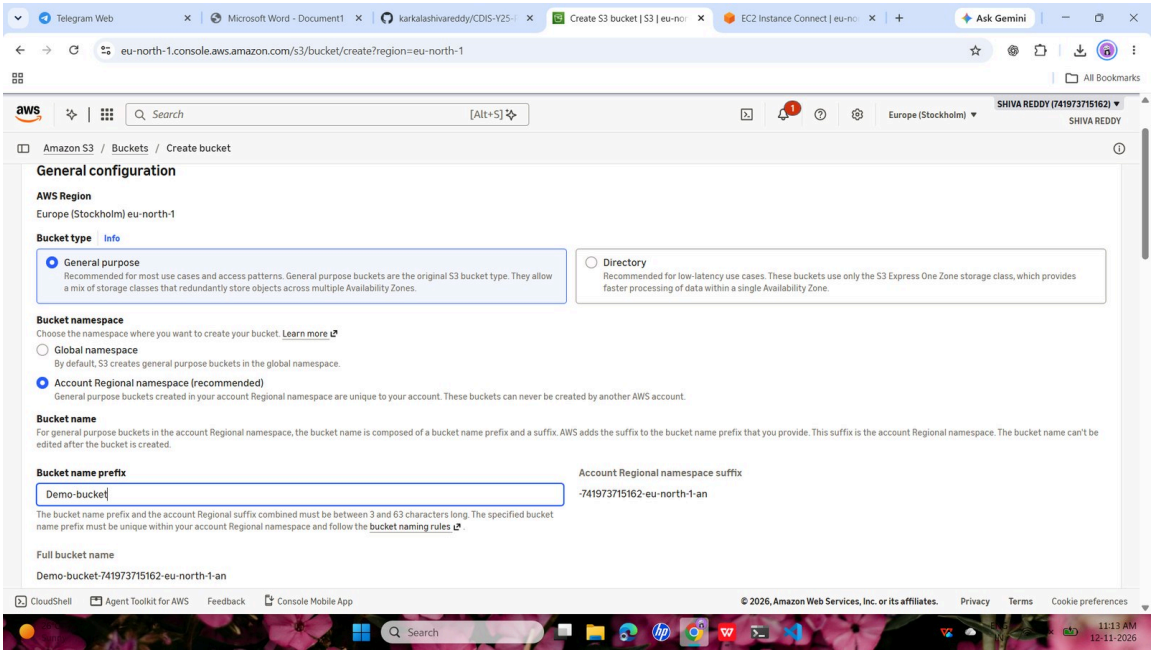


Fig 17: Create bucket - entering the bucket name

In the Create bucket page the Account Regional namespace was selected and the prefix Demo-bucket was typed first. The name was later changed to shiva-s3-bucket before the bucket was created.

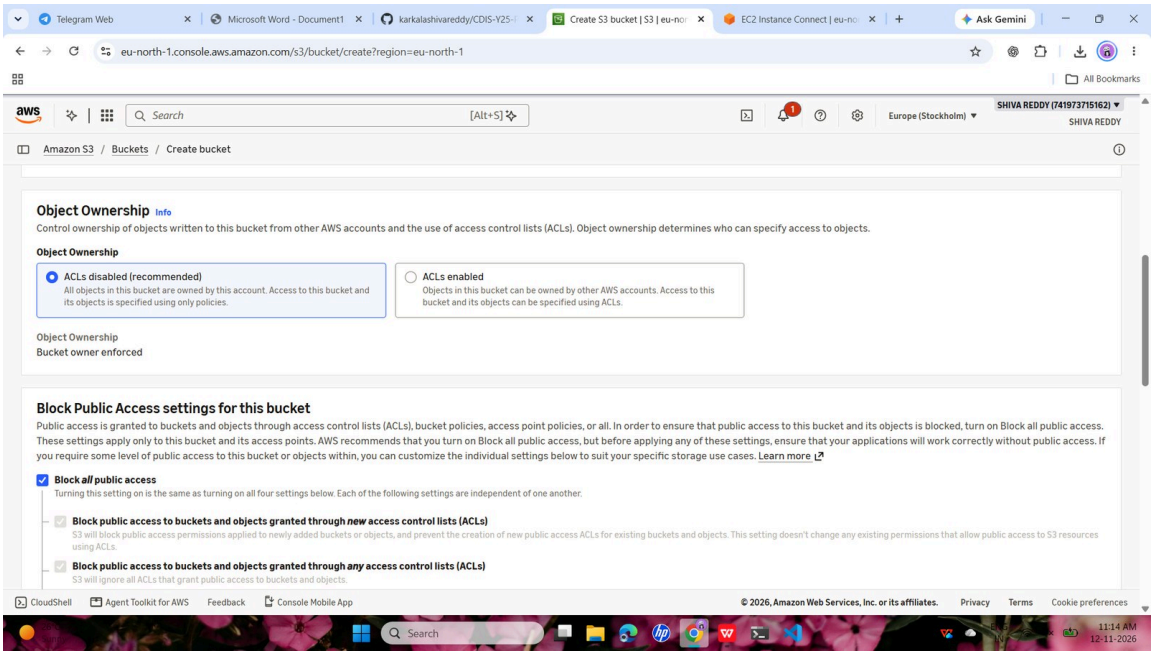


Fig 18: Object ownership and public access settings

ACLs were kept disabled so that the bucket owner owns all the objects. Block all public access was kept enabled to keep the bucket and its objects private.

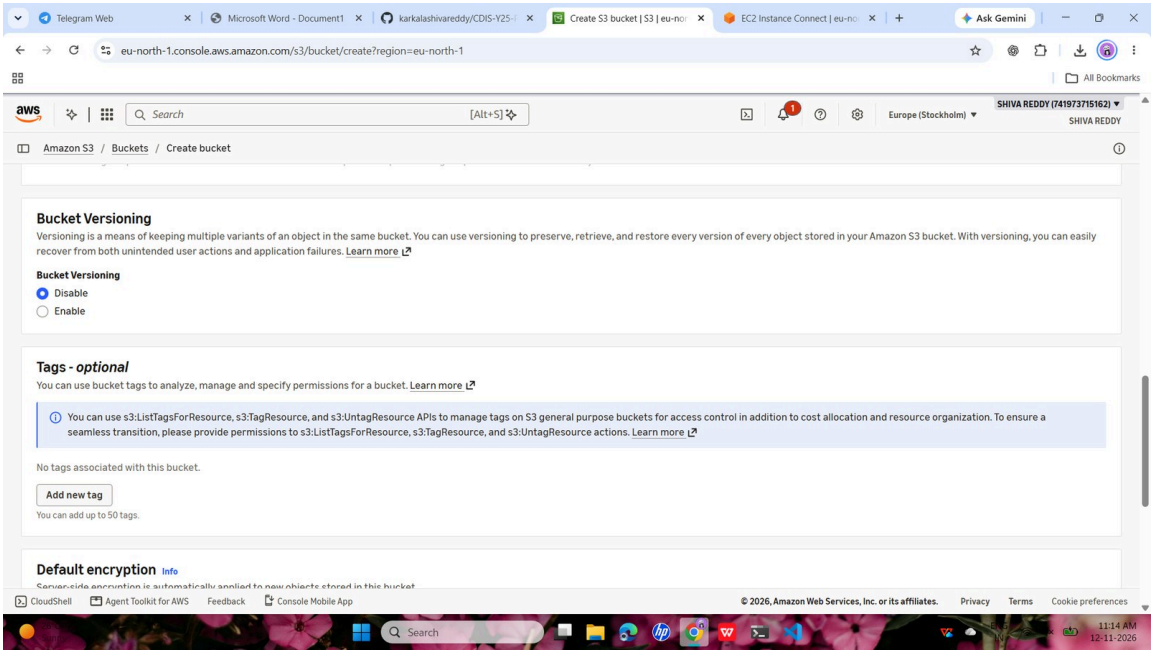


Fig 19: Bucket versioning and tags

Bucket versioning was left disabled and no tags were added. These are the default settings, and the bucket was created after the remaining options were reviewed.

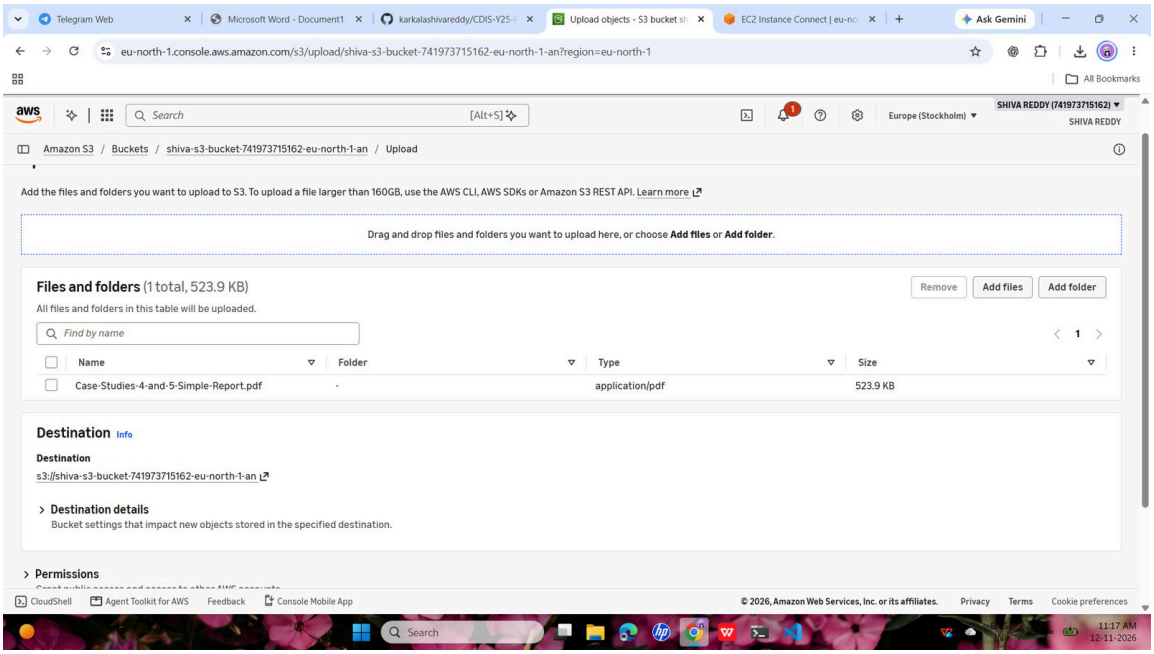


Fig 20: Selecting files for upload

The Upload page was opened for the new bucket. The first file, a PDF report, was selected using Add files and the destination was shown as the bucket URI. The remaining files with other extensions were added in the same way.

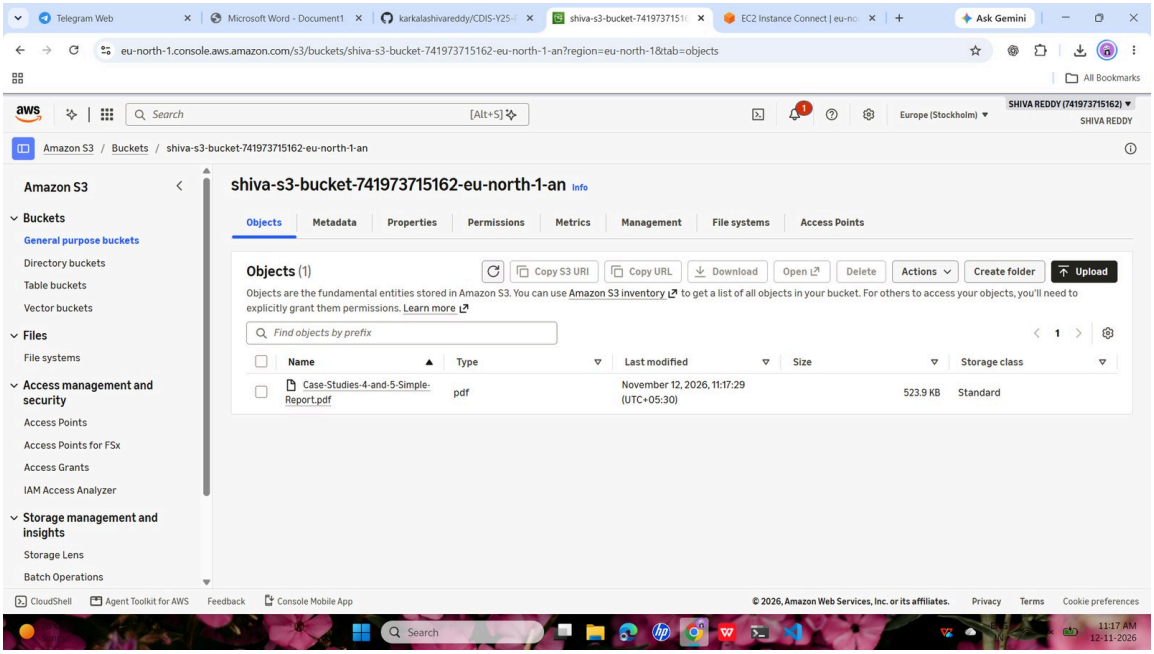


Fig 21: Uploaded object in the bucket

After the upload, the PDF appeared in the Objects tab with its size, last modified time and Standard storage class. The remaining four files were uploaded in the same way, giving five objects with different extensions.

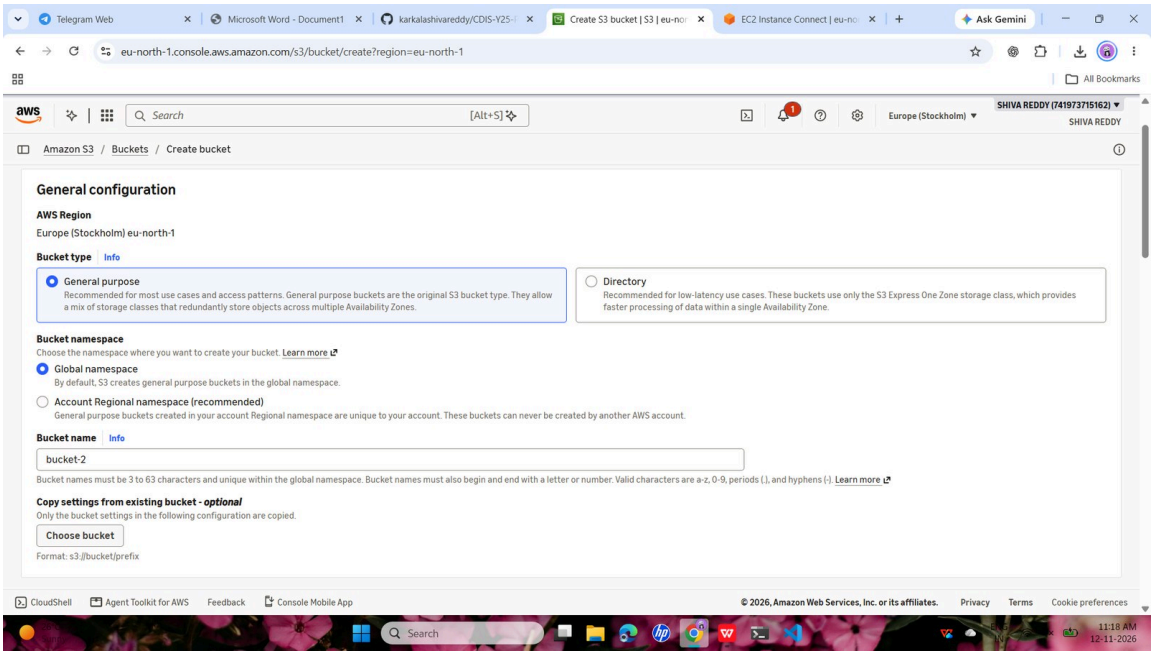


Fig 22: Creating a second bucket

A second bucket was created to practise unique bucket naming. The Global namespace option was selected and a temporary name was entered, which was changed to shiva-s3-bucket1 before creation.

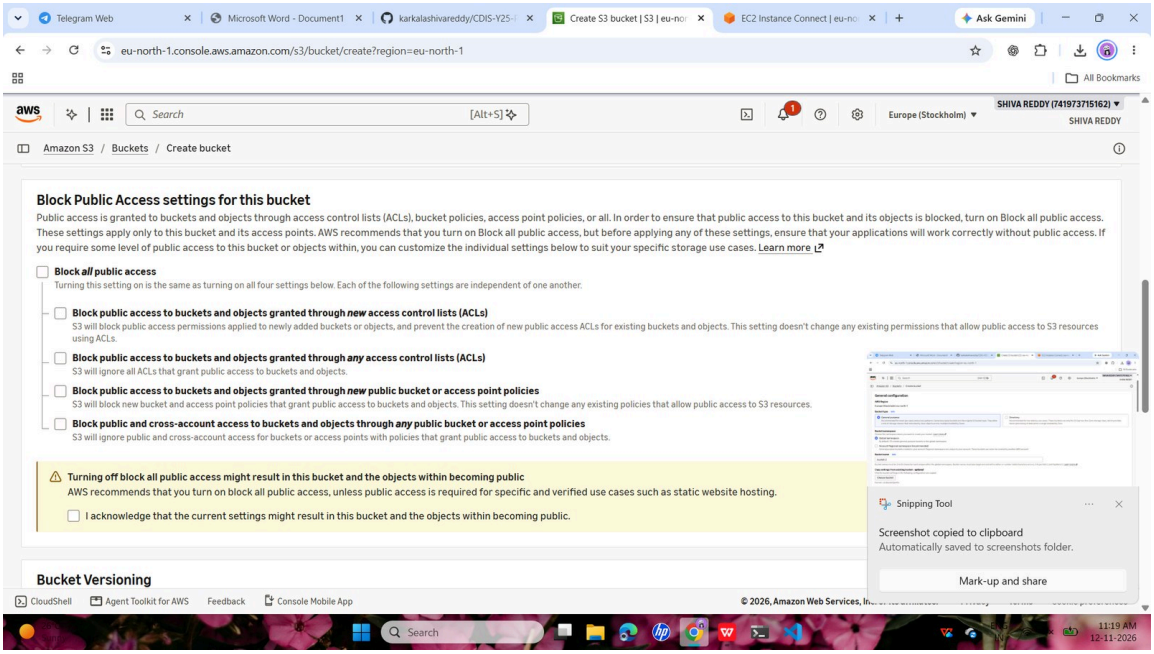


Fig 23: Block public access settings of the second bucket

The Block Public Access options were reviewed before creating the second bucket. The settings were left unchecked here, and the warning explains that turning off block all public access can make the bucket and its objects public.

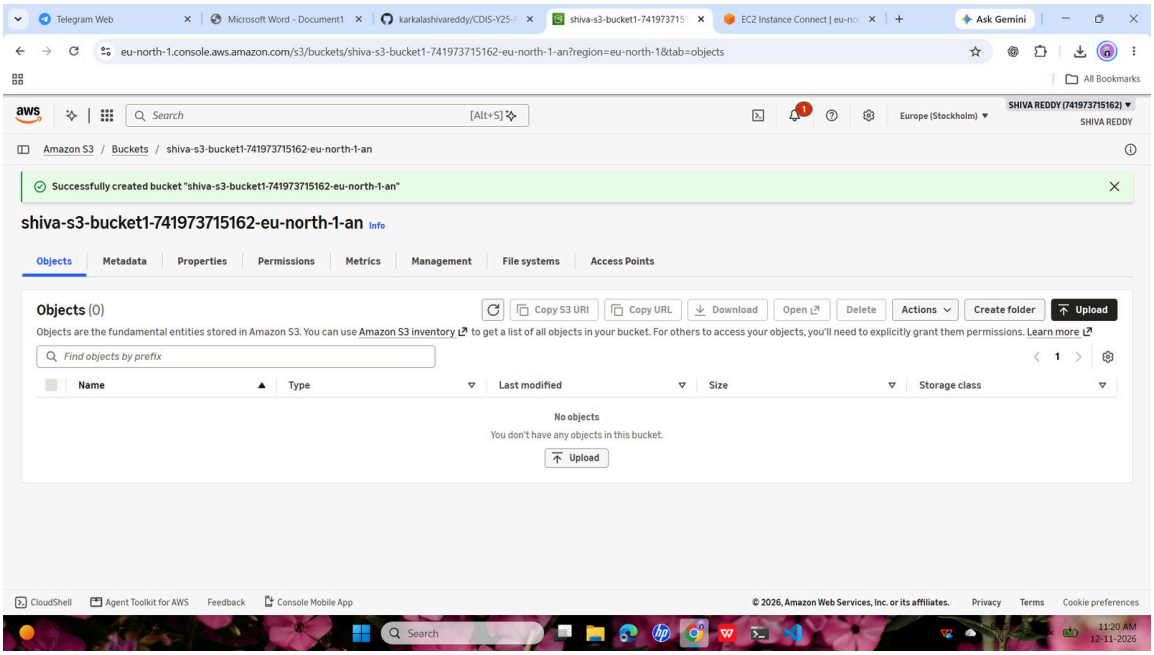


Fig 24: Second bucket created successfully

The second bucket shiva-s3-bucket1-741973715162-eu-north-1-an was created in the Europe (Stockholm) region. Its Objects tab was empty at this point and the Upload button was available.

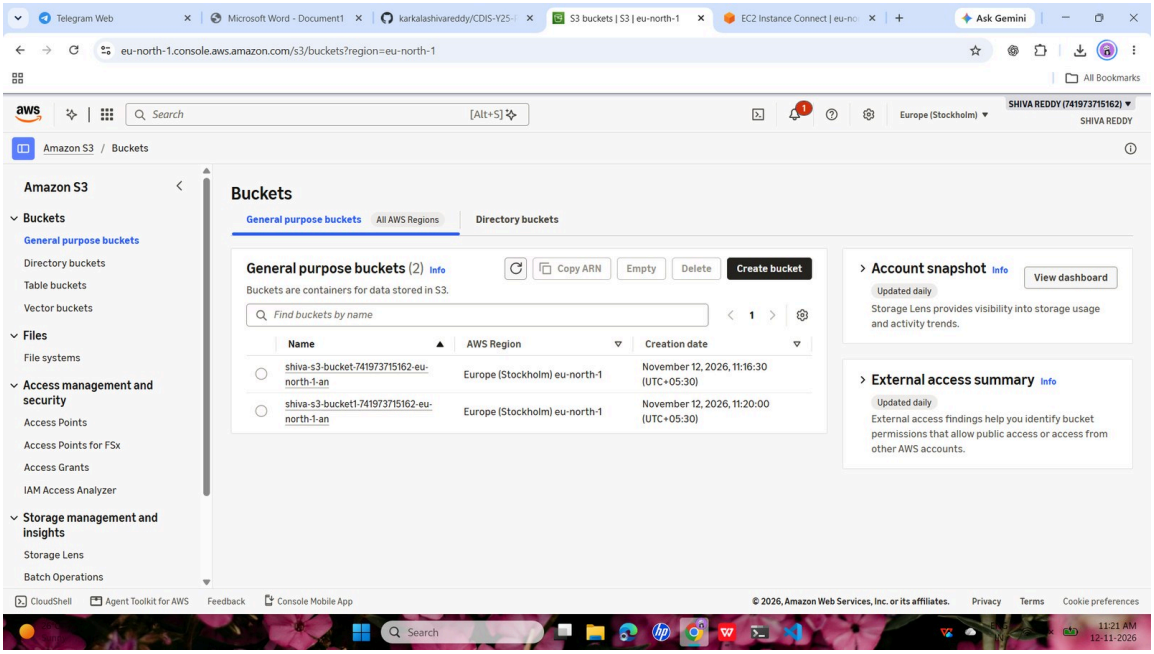


Fig 25: List of buckets in the account

The Buckets page listed both general purpose buckets created in this practical, with their region and creation time. This confirmed that the buckets were available for storing objects.